

Multi-State EV Site Suitability Analysis

A reusable ArcGIS Experience Builder methodology for federal EV charging infrastructure eligibility, deployed independently for Tennessee, Oklahoma, and Mississippi's NEVI programs

The Problem

Federal EV infrastructure funding programs run independently in each state under the same national NEVI Formula Program, but each state DOT needs to translate the same core federal eligibility rules, corridor proximity, charger spacing, equity criteria, into its own spatial reality: its own designated corridors, its own utility territories, its own existing charger inventory. Building separate one-off applications from scratch for each state would mean re-solving the same core spatial analysis problem, network-based eligibility zones, corridor gap analysis, multi-criteria layering, several times over, with several separate chances to get the underlying methodology inconsistent between states.

- Every state's designated Alternative Fuel Corridors differ in number, length, and route, but the eligibility rule applied to them is identical: within a set travel distance of a qualifying interchange.
- Coverage gap analysis needs to be repeatable and defensible in the same way across states, so a state's determination of where funding priority should go isn't built on a different, less rigorous analysis than a neighboring state used.
- Each state layers in its own supporting data, its own utility territories, its own reporting cadence, but a consistent underlying spatial methodology matters more than any individual state's specific presentation choices.
- These applications serve non-GIS audiences directly, developers, utility staff, and the public, so the interaction design has to be intuitive and self-explanatory in every state's deployment, not just usable to someone who already understands the underlying analysis.

My Approach

I built and deployed a reusable ArcGIS Experience Builder pattern for EV site suitability, applied independently across Tennessee, Oklahoma, and Mississippi's NEVI programs, adapting each deployment to that state's specific corridors and data while keeping the underlying spatial methodology consistent.

One core spatial method, reused everywhere. Every state's eligibility zone around a qualifying interchange is built the same way, a network drive-distance analysis along the actual public road network, not a circular buffer, so the boundary in every state means the same thing: the true travel distance a vehicle would need to cover. The same logic extends to any state with a designated non-limited-access corridor, where the corridor gets

segmented to enforce the required maximum spacing directly in the map geometry rather than as a separate written rule.

A consistent gap analysis method with state-specific compliance data. Each state's existing charger inventory, sourced from a mix of federal and state verification data, feeds the same underlying gap detection logic: identify any stretch of a designated corridor that exceeds the program's spacing limit without a station meeting the required simultaneous fast-charging output. What changes state to state is the input data; what stays constant is the analysis.

A common interaction pattern adapted per state's data. Corridor filtering, utility and equity layer toggles, feature pop-ups with the specific attributes a developer needs to act on, these interaction patterns repeat across all three deployments, so the learning curve for a user familiar with one state's application transfers almost entirely to the next. Each state's specific data, utility territory boundaries, funding round structure, downloadable file formats, gets layered onto that same interaction foundation rather than reinventing it.

Purpose-built for non-GIS users in every deployment. None of these applications assume the person using them has any GIS background. Every eligibility boundary, gap zone, and funding status is presented as something to look at and understand immediately, corridor filters instead of query builders, plain-language pop-ups instead of raw attribute tables, so a site developer or a member of the public gets the same clarity a GIS analyst would.

Technical Highlights

- **Platform:** ArcGIS Experience Builder, deployed independently for Tennessee, Oklahoma, and Mississippi's NEVI programs on each state's ArcGIS Online organization
- **Data sources:** FHWA alternative fuel corridor designations, state DOT corridor and interchange data, US Department of Energy Alternative Fuels Data Center, state and federal equity screening data, state utility territory data
- **Spatial analysis:** Network drive-distance polygons for interchange eligibility zones, corridor-wide gap analysis against each program's spacing and power output requirements, applied consistently across all three deployments
- **Interactivity:** Corridor and status filtering, toggleable utility and equity layers, geocoded search, detailed feature pop-ups, direct data download where applicable
- **Scope:** Three statewide deployments covering federally designated alternative fuel corridors across Tennessee, Oklahoma, and Mississippi, supporting a combined federal allocation exceeding \$200 million

Why It Matters

A reusable methodology across state lines means each new deployment builds on validated analysis logic rather than starting from a blank page, and it means a developer or utility

working across state boundaries encounters a consistent, predictable tool rather than three unrelated ones. Applying the same network-distance eligibility standard, the same gap detection logic, and the same non-GIS-first interaction design in every state kept each application's underlying spatial analysis defensible and kept the build-out effort proportional to what actually changes state to state, the corridors and the data, not the fundamentals of how the analysis works.